

In-gel protein staining with Coomassie Brilliant Blue

Coomassie Brilliant Blue is a pigment used to stain proteins in acrylamide or agarose gels with a detection limit of 10–20 ng per band. The staining is permanent and incompatible with Western blotting. However, Coomassie G-250 (not R-250) can also be used to visualize proteins on PVDF or nitrocellulose membranes. While silver staining [SOP0011](#) offers higher sensitivity, Coomassie-stained gels are compatible with downstream mass spectrometry.

Risk assessment

- Work with flammable liquids
- Work with corrosive liquids
- ▷ Wear gloves, safety glasses, lab coat
- Collect as HAZARDOUS WASTE



Reviewed: Jan 16, 2023

Procedures

>> Coomassie Brilliant Blue staining

- | | |
|--|--|
| ☐ Incubation tray | ☐ Coomassie destaining solution (20 mL / 1 rxn) R0123 |
| ☐ Lint-free wipe | ☐ Coomassie staining solution (20 mL / 1 rxn) R0124 |

- (1.) *Optional:* Place gel in a container with water. Microwave at 1 000 W for 1 min.



This is why: This helps remove residual SDS and speeds up destaining. Don't boil; gels can melt or crack if microwaved dry.

- (2.) Transfer gel into a tray with Coomassie staining solution. Rock gently at room temperature for 15 min.

⌚ 15 min

- (3.) Discard staining solution. Rinse briefly with water to remove excess dye.



- (4.) Add Coomassie destaining solution. Rock at room temperature for 45 min.

⌚ 45 min

- (5.) Pour off the destaining solution into a collection bottle for reuse. Replace with water and add a lint-free wipe in the tray to soak up free dye. Destain overnight until background disappears.



Materials

Coomassie Brilliant Blue staining

Incubation tray

Lint-free wipe

Coomassie destaining solution [R0123](#) (20 mL / 1 rxn)

Coomassie staining solution [R0124](#) (20 mL / 1 rxn)

Recipes

Coomassie destaining solution

Amount	Ingredient	Stock	Final
1.2 L	Isopropyl alcohol Flammable liquid (H225)	[67-63-0] 100%	30%
2.4 L	Water, reagent-grade	[7732-18-5]	
0.4 L	Acetic acid, glacial Skin corrosion (H314)	[64-19-7] 99.6%	10%

Add concentrated acid slowly to the stirred water/alcohol mixture. DO NOT autoclave. **Safety:** Work inside a certified chemical fume hood. **Note:** The destaining solution can be reused several times and refreshed using activated charcoal. **This is why:** In this adaptation, 30% isopropyl alcohol or ethanol replaces the more toxic 45% methanol.

Coomassie destaining solution

30% Isopropyl alcohol, 10% Acetic acid



DANGER

Flammable liquid; Skin and serious eye irritation; Drowsiness or dizziness

- ☐ Collect as ORGANIC SOLVENT WASTE
- ☐ Keep away from oxidizers

Date: Sign: R0123

Coomassie staining solution

Amount	Ingredient	Stock	Final
1.0 g	Brilliant Blue G	[6104-58-1] 10 g/L	0.5 g/L
2.0 L	Coomassie destaining solution Flammable liquid (H225); Skin irritation (H315); Serious eye irritation (H319)	↗ R0123	

Note: The staining solution can be used multiple times. **Hint:** Coomassie G-250 is more sensitive while Coomassie R-250 gives sharper bands at lower cost.

Coomassie staining solution



DANGER

Flammable liquid; Skin and serious eye irritation; Drowsiness or dizziness

- ☐ Collect as ORGANIC SOLVENT WASTE
- ☐ Keep away from oxidizers

Date: Sign: R0124

Troubleshooting

Coomassie Brilliant Blue staining

In Step 2:

- Faint or invisible bands
 - Increase staining time or protein loading. Ensure sufficient protein was loaded. The detection limit is approximately 10–20 ng per band.
 - Verify staining solution has not been exhausted from repeated reuse.
- Uneven staining across the gel
 - Ensure the gel is fully submerged and rocking evenly. Use sufficient volume of staining solution — at least 5 times the gel volume.

In Step 4:

- High background that does not clear with extended destaining
 - Replace destaining solution more frequently. Add a crumpled lint-free wipe or activated charcoal to absorb free dye.
 - If using R-250, switch to G-250, which gives lower background due to colloidal binding.

Change log

2020-06-12	Hai T. Dao	Initial protocol; Tom Muir lab "Coomassie stain and destain solution".
2023-01-16	Benjamin C. Buchmuller	Adaptation as SOP. Replaced methanol with less toxic isopropyl alcohol, from the Daniel Summerer lab "Coomassie staining" protocol.

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